ISYE 6740 Final Project

Course: ISYE 6740

Group: Group #285

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Project Title: Graph-Based Spectral Analysis of Paddy Farming Practices

Graph-Based Spectral Analysis of Paddy Farming Practices

1. Introduction

Agricultural production depends on a combination of farm size, crop variety, soil type, fertilizer application, weather, irrigation, and other cultivation practices. In this project, the UCI Paddy Dataset was analyzed to identify groups of farms with similar farming profiles and to detect farms that deviate from typical patterns. The central question was whether farms could be represented as a similarity network and then grouped into interpretable clusters of farming practices.

The project began with a graph-based spectral clustering proposal. During implementation, the analysis was expanded to include exploratory data analysis, preprocessing, PCA dimensionality reduction, graph construction, spectral clustering, baseline clustering comparisons, cluster interpretation, anomaly detection, and a web portal. During evaluation, spectral clustering was compared with simpler baseline methods. The final interpretation uses k-means clustering because it produced the strongest internal validation scores, while spectral clustering remains an important graph-based comparison method.

2. Dataset

The dataset contains 2,789 farm records and 45 original variables. The variables include farm characteristics, agriblock, crop variety, soil type, nursery practice, seed rate, fertilizer applications, pesticide usage, rainfall, irrigation, temperature, wind, relative humidity, trash bundles, and paddy yield. In this context, Agriblock refers to the agricultural block or local farming region where the observation was recorded. The dataset includes six agriblocks: Chinnasalem, Cuddalore, Kallakurichi, Kurinjipadi, Panruti, and Sankarapuram.

The dataset had no missing values. However, the preprocessing pipeline still includes imputation as a safeguard for future data or portal uploads. The initial EDA found 451 duplicate rows beyond the first occurrence, which is relevant because duplicate or repeated farm profiles can influence PCA, graph construction, and clustering.

The target-like outcome column, Paddy yield(in Kg), was excluded from clustering features. It was preserved separately and used only after clustering to interpret whether the discovered groups were meaningful.

3. Preprocessing

The preprocessing workflow cleaned column names, separated numeric and

categorical variables, standardised numeric features, and one-hot encoded

categorical features. Standardisation was important because the numeric

variables were measured on different scales, such as hectares, kilograms,

tonnes, millimetres, millilitres, and temperature. Without standardisation,

variables with larger raw magnitudes could dominate Euclidean distances and

distort PCA or clustering results.

The original feature set used for clustering contained 44 input columns after

excluding yield. Of these, 36 were numeric and 8 were categorical. The

categorical columns produced 35 one-hot encoded columns, resulting in 71

processed feature columns.

The categorical variables included:

```
Feature  |  Known categories

Agriculture  |  Y

Variety  |  Y

Soil Type  |  Y

Nursery  |  Y

Area Protection_NO_PEST  |  Y

Area Protection_YES_PEST  |  Y

Area Protection_NO_PEST  |  Y

Area Protection_YES_PEST  |  Y
```

4. PCA Dimensionality Reduction

Principal Component Analysis was applied to the 71 processed features. PCA

showed that the data could be reduced substantially while retaining most of

the variance.

Key PCA results:

```
Component summary  |  Value

PC1 explained variance  |  26.95%

PC2 explained variance  |  20.40%

PC1 + PC2 cumulative variance  |  47.34%

Components needed for 95% variance  |  31
```

The first two components were used for visualisation, while the first six

components were used for graph construction, clustering, and anomaly

detection. This reduced noise and avoided performing distance-based

clustering in the full 71-dimensional encoded feature space. The cumulative

explained variance plot shows that the first six components preserve at least

80% of the variance, making six components a reasonable modelling choice.

PCA cumulative explained variance

The two-dimensional PCA projection also shows a structured pattern rather

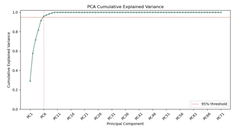
than a random cloud. When the points are colored by yield, yield varies

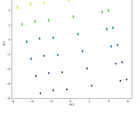
strongly across the projection, indicating that some PCA directions are

closely related to farm scale and input intensity.

PCA projection colored by paddy yield

5. Similarity Graph Construction





A similarity graph was constructed using the first six PCA components. Each

form was represented as a node. Edges were created using a k-nearest-neighbor

approach, and edge weights were computed with a Gaussian similarity function

based on Euclidean distance in PCA space. The purpose of the graph was to

represent forms as a network of similar observations, so that graph-based

spectral clustering could use the connectivity structure rather than only

direct point-to-point distances.

Smaller nearest-neighbor values produced highly disconnected graphs. For

example, k=10 produced 147 connected components. The graph became fully

connected at k=600, so that value was used for the final similarity graph.

Final graph summary:

```
Metric | Value

Nodes | 2,780

Edges | 833,400

Nearest neighbors | 600

PCA components used | 6

Connected components | 1

Median degree | 600

Mean degree | 602.76
```

This graph was used for spectral clustering and graph-based anomaly features.

The graph overlay on the PCA projection gives a visual sense of how forms are

connected in the reduced feature space.

6. Clustering Methods

Three clustering approaches were compared:

1. Spectral clustering using the similarity graph

2. K-means clustering on the six PCA components

3. Hierarchical Ward clustering on the six PCA components

Cluster counts from k=2 through k=10 were evaluated. The metrics used were

silhouette score, Calinski-Harabasz score, and Davies-Bouldin score.

Best results by method:

```
Method  | k | Silhouette | Calinski-Harabasz | Davies-Bouldin

k-means | 10 | 0.3486 | 1867.84 | 1.0655

spectral | 10 | 0.3486 | 1861.57 | 0.7981

hierarchical_ward | 10 | 0.3458 | 1865.61 | 0.7955
```

K-means slightly outperformed spectral clustering across the internal

validation metrics. Since the difference was not large, this suggests that

both k-means and spectral clustering found similar structure in the PCA-

reduced space. K-means was selected as the final model for cluster

interpretation because it had the best validation scores.

The silhouette comparison plot shows that all three methods improved as the

number of clusters increased within the tested range. Since the best score

occurred at k=10, future work could test larger values such as k=11 through

k=15. For this project, however, k=10 was kept as the final cluster count

because it was the best value within the planned evaluation range.

7. Cluster interpretation

The final k-means model used k=10. Cluster profiling showed that the

discovered groups are strongly associated with farm/input scale and

agrilblock. Higher-yield clusters generally had larger average hectares and

larger input quantities.

Cluster summary

| Cluster | Farms | Avg yield kg | Avg hectares | Dominant agrilblock | Dominant variety

| Dominant soil | Dominant scenery

| 0 | 380 | 26,363.07 | 4.76 | Gokwanyane | Wheat penny | Crop | Dry

| 1 | 286 | 29,535.07 | 4.76 | Nyirolopoko | Lemmer | Crop | Dry

| 2 | 266 | 26,151.56 | 4.76 | Ratlhatlosito | Lemmer | Crop | Dry

| 3 | 266 | 26,196.61 | 4.76 | Tlakgotone | White potato | Crop | Dry

| 4 | 333 | 81,496.97 | 4.85 | Chinsekelak | Lemmer | Crop | Dry

| 5 | 405 | 32,328.67 | 3.89 | Mareetsi | White penny | Crop | Dry

| 6 | 237 | 13,833.80 | 2.27 | Tshelanyemb | White potato | Crop | Dry

| 7 | 287 | 32,764.92 | 2.21 | Nyirolopoko | Lemmer | Crop | Dry

| 8 | 286 | 32,957.76 | 2.26 | Tlakgotone | White potato | Crop | Dry

| 9 | 270 | 51,254.31 | 1.97 | Ratlhatlosito | Lemmer | Afrisoil | Dry

High-yield clusters included clusters 8, 9, 1, and 7. These had average

yields around 29,100 to 29,450 kg and average farm sizes around 4.76

hectares. Low-yield clusters included clusters 3, 0, 6, and 8. These had

average yields around 13,300 to 13,000 kg and average farm sizes around 2

hectares.

This indicates that the clustering model is capturing meaningful differences

in farm scale, input intensity, and regional farming profiles. The clusters

are not arbitrary labels; they correspond to distinct combinations of

agrilblock, farm size, input levels, and yield outcomes.

The average-yield chart makes the scale separation especially clear. Clusters

with larger farm area and higher input levels also tend to have higher total

yield.

The cluster profile heatmap standardizes key numeric variables across

clusters. Positive values indicate a cluster is above the overall average for

that variable, while negative values indicate it is below the overall

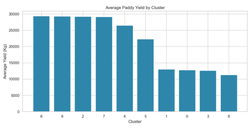
average. This helps show that high-yield clusters are also generally higher

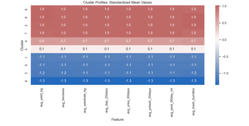
in hectares, seed rate, fertilizer input, pesticide use, and trash bundles.

The yield boxplot shows the spread of yield within each cluster and helps

confirm that cluster-level yield differences are not only driven by single

extreme values.





8. Anomaly Detection

Anomaly detection combined three unsupervised signals:

Signal | Weight

Distance from assigned cluster centroid | 45%

Isolation Forest anomaly score | 45%

Low-weighted graph degree | 10%

The top 5% most anomalous farms were flagged, producing 139 top anomalies.

Cluster 8 had the largest number of top anomalies, with 38 farms in the top

5%. Clusters 6 and 9 had zero top-5% anomalies, suggesting they were more

internally consistent.

Example top anomaly:

Field | Value

Node | 2786

Cluster | 8

Hectares | 1

Agriblock | Chinnasalem

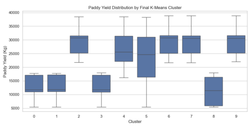
Variety | delux ponni

Soil type | clay

Seedrate | 25 kg

Nursery | wet

Paddy yield | 5,723 kg



combined anomaly score | 0.0618

The strongest anomalies were mostly very small one-hectare farms with low

seed rates, low fertiliser inputs, and low yields. These farms were far from

their assigned cluster centers and were also flagged by Isolation Forest.

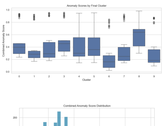
Cluster 4 had the highest concentration of top anomalies, which is consistent

with its lower average hectares and lower average yield.

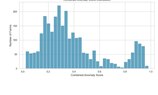
Anomalies by cluster

The anomaly score distribution shows that most farms have moderate anomaly

scores, while a smaller group of farms separates into the high-anomaly tail.



Combined anomaly score distribution



9. Web Portal

A static web portal was created in the app/ directory. The portal displays

the major project outputs, including PCA projections, cluster profiles,

clustering model comparisons, graph connectivity, and anomaly results. The

portal was designed to make the analysis easier to inspect interactively,

especially for comparing clusters and reviewing anomalous farms.

The portal can be opened directly from

app/index.html

No server or additional Python web framework is required.

10. Limitations

The analysis is unsupervised, so there are no true cluster labels to validate

against. Internal validation metrics are useful, but they do not prove that a

clustering result is agronomically optimal.

The dataset also contains repeated or duplicate rows. These repeated profiles

may reflect real repeated farm conditions, data collection patterns, or

duplication artifacts. They should be discussed as a limitation because they

can affect similarity graph structure and cluster compactness.

Yield appears highly related to farm size and input quantities. Therefore,

the clusters may primarily reflect scale and input intensity rather than

purely qualitative differences in farming strategy. A possible future

extension would normalize yield by hectares and analyze yield efficiency

instead of total yield

11. Conclusion

This project identified meaningful structure in the UCI Paddy Dataset using

preprocessing, PCA, graph construction, clustering, baseline comparison,

cluster interpretation, and anomaly detection.

Although spectral clustering was the original graph-based method, k-means

slightly outperformed spectral clustering on internal validation metrics. The

final k-means model with 10 clusters revealed interpretable farm groups

driven mainly by agritech, farm size, input levels, and yield. Anomaly

detection further identified farms with unusual low-scale, low-input, and

low-yield profiles.

Overall, the project shows that unsupervised learning can uncover useful

patterns in paddy farming practices and can support exploratory agricultural

analysis through cluster profiling and anomaly detection.